

# Coordinate Geometry (Set B) — MCQ Worksheet

**Mathematics** • Chapter 7 • 25 Questions, 1 mark each — Total: 25 marks

**Instructions:** Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

**Q1.** Distance between (2, 3) and (5, 7) is:

- (A) 5
- (B) 4
- (C) 3
- (D)  $\sqrt{50}$

**Q2.** Distance of point (−3, 4) from origin is:

- (A) 7
- (B) 5
- (C) 1
- (D) −5

**Q3.** Midpoint of (4, 6) and (−2, 2) is:

- (A) (1, 4)
- (B) (2, 4)
- (C) (1, 8)
- (D) (3, 4)

**Q4.** The point dividing segment joining (1, 2) and (4, 8) internally in ratio 1:2 is:

- (A) (2, 4)
- (B) (3, 6)
- (C) (5/3, 4)
- (D) (2, 6)

**Q5.** Are (1, 5), (2, 3) and (−2, 11) collinear?

- (A) Yes
- (B) No
- (C) Only sometimes
- (D) Cannot say

**Q6.** Area of triangle with vertices (1, 1), (4, 1), (1, 5):

- (A) 6 sq units
- (B) 12 sq units
- (C) 5 sq units
- (D) 10 sq units

**Q7.** Centroid of triangle with vertices (0, 0), (6, 0), (0, 9):

- (A) (2, 3)
- (B) (3, 3)
- (C) (2, 2)
- (D) (3, 2)

**Q8.** The reflection of (4, −3) in the origin is:

- (A) (−4, 3)
- (B) (4, 3)
- (C) (−4, −3)
- (D) (3, −4)

**Q9.** Distance between (a, 0) and (0, b) is:

- (A)  $a + b$
- (B)  $a - b$
- (C)  $\sqrt{a^2 + b^2}$
- (D)  $ab$

**Q10.** If P(2, 3) divides AB internally where A = (1, 2), B = (4, 5), then AP:PB =

- (A) 1:2
- (B) 2:1
- (C) 1:1
- (D) 1:3

**Q11.** Which axis is the equation  $x = 0$ ?

- (A) x-axis
- (B) y-axis
- (C) Both
- (D) Neither

**Q12.** In which quadrant does point (3, −2) lie?

- (A) First
- (B) Second
- (C) Third
- (D) Fourth

**Q13.** If the points (a, 0), (0, b) and (1, 1) are collinear, then  $1/a + 1/b =$

- (A) 0
- (B) 1
- (C) −1
- (D) 2

**Q14.** Distance between  $(\cos\theta, \sin\theta)$  and  $(-\sin\theta, \cos\theta)$ :

- (A) 1
- (B) 2
- (C)  $\sqrt{2}$
- (D)  $\sqrt{3}$

**Q15.** Find ratio in which the segment joining (1, −5) and (−4, 5) is divided by x-axis:

- (A) 1:1  
(C) 2:3

- (B) 1:2  
(D) 3:5

**Q16.** The vertices of a triangle are (1, 0), (−1, 0), (0, 2). Triangle is:

- (A) Equilateral  
(C) Isosceles

- (B) Right-angled isosceles  
(D) Scalene

**Q17.** Find the coordinates of the point on x-axis equidistant from (−2, 5) and (2, −3).

- (A) (−2, 0)  
(C) (0, 0)

- (B) (2, 0)  
(D) (1, 0)

**Q18.** If A(−1, 3) and B(3, −1) are endpoints of a diameter of circle, the centre is:

- (A) (1, 1)  
(C) (2, 0)

- (B) (0, 2)  
(D) (1, 2)

**Q19.** Find the value of k if points (k, 2), (−1, 1), (1, 3) are collinear.

- (A)  $k = 1$   
(C)  $k = 0$

- (B)  $k = -1$   
(D)  $k = 2$

**Q20.** Three vertices of a parallelogram ABCD are A(1, 2), B(4, 3), C(6, 6). Find D.

- (A) (3, 5)  
(C) (3, 4)

- (B) (2, 4)  
(D) (5, 5)

**Q21.** The area of a triangle is 5 sq units. Two of its vertices are (2, 1), (3, −2). The third vertex lies on  $y = x + 3$ . Find third vertex.

- (A)  $(7/2, 13/2)$  or  $(-3/2, 3/2)$   
(C) (3, 6)

- (B) (2, 5)  
(D) (1, 4)

**Q22.** If point P(x, y) is equidistant from A(7, 1) and B(3, 5), prove the relation between x and y:

- (A)  $x - y = 2$   
(C)  $x - y = 4$

- (B)  $x + y = 2$   
(D)  $x + y = 4$

**Q23.** Find the length of median from vertex A to side BC in triangle with A(5, 1), B(1, 5), C(−3, −1):

- (A) 5 units  
(C)  $\sqrt{37}$  units

- (B) 6 units  
(D)  $5\sqrt{2}$  units

**Q24.** If A(2, 0), B(0, 2) and C(0, 7) are three vertices of a trapezium ABCD with  $AB \parallel CD$ , then D =

- (A) (5, 0)  
(C) (0, 5)

- (B) (7, 0)  
(D) (0, 7)

**Q25.** The points (3, 2), (−1, −3) and (5, 6) form a triangle whose area is:

- (A) 12.5 sq units  
(C) 1.5 sq units

- (B) 0.5 sq units  
(D) 2.5 sq units